Properties of Quadratic Functions

For

$$f(x) = ax^2 + bx + c, \qquad a \neq 0,$$

the graph is a parabola with a vertical axis of symmetry.

mx+b, (0,b) y-...

Feature	Formula or condition	Meaning
Opening	$a > 0$ or $a < 0$	up for $a > 0$; down for $a < 0$
Axis of symmetry	$x = -\frac{b}{2a}$	vertical line through the vertex
Vertex	$\left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right)$	minimum if $a > 0$; maximum if $a < 0$
y-intercept	$(0, c)$	value of the function at $x = 0$

- Quadratic function is x^2 after transformation.

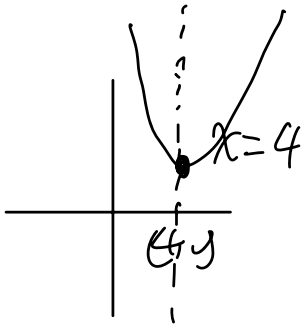
  ∪, ∩, ⋃, ⌒, ...

- $a>0$, ∪ $a<0$, ∩ $a\neq 0$. $f(x)=ax^2+bx+c$ $=bx+c$ /

- ∪ ← axis of symmetry

  $x=\frac{-b}{2a}$.

  "ex" $x=4$



- ∩ ∪

  $\left(-\frac{b}{2a}, f\left(\frac{-b}{2a}\right)\right)$

  ex: $-\frac{b}{2a}=4$

  $(4, f(4))$

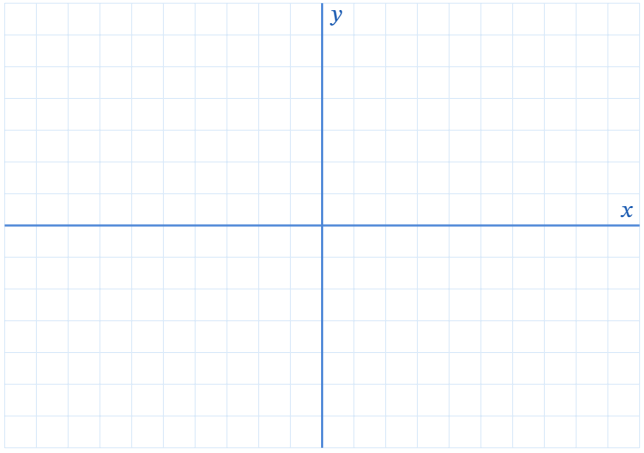
The Squaring Function

The elementary quadratic function is

$$f(x) = x^2.$$

- vertex: $(0, 0)$;
- axis of symmetry: $x = 0$;
- domain: $(-\infty, \infty)$;
- range: $[0, \infty)$;
- minimum value: 0;

Plot at least five points and sketch the parabola.



Find the Vertex from Standard Form

For $f(x) = ax^2 + bx + c$, first find $\left(-\frac{b}{2a}, f\left(\frac{-b}{2a}\right)\right)$

Then the vertex is and the axis of symmetry is $x = \frac{-b}{2a}$

Find the vertex and axis of symmetry of each function.

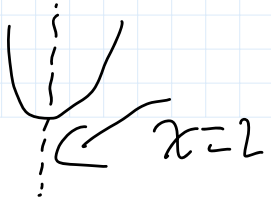
$f(x) = x^2 - 4x + 5$

$f(2) = 2^2 - 8 + 5 = 1$

- axis of symmetric.

$a=1, b=-4, c=5.$

$a>0$,



$\frac{-b}{2a} = \frac{-(-4)}{2 \cdot 1} = \frac{4}{2} = 2$

$x=2$ is axis of symmetry

- vertex. $\left(\frac{-b}{2a}, f\left(\frac{-b}{2a}\right)\right)$

$(2, f(2)) = (2, 1)$

$g(x) = x^2 - x - 6$

- $a=1, b=-1, c=-6$
- $\frac{-b}{2a} = \frac{1}{2}$, $x = \frac{1}{2}$
- $\left(\frac{1}{2}, f\left(\frac{1}{2}\right)\right)$

$= \left(\frac{1}{2}, -\frac{25}{4}\right)$

$\left(\frac{1}{2}\right)^2 - \frac{1}{2} - 6$

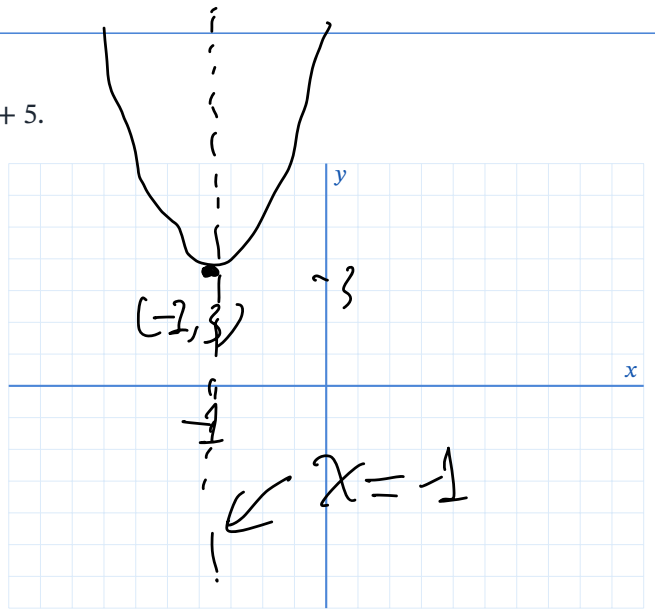
$= -\frac{25}{4}$

Graph from Standard Form

Graph $f(x) = 2x^2 + 4x + 5.$

$\left(\frac{-b}{2a}, f\left(\frac{-b}{2a}\right)\right)$

$= (-1, 3)$



State each feature.

- opening direction; $a=2$, $a>0$, $\cup$
- vertex; $(-1,3)$
- axis of symmetry; $x=-1$
- y-intercept; $(0,c)=(0,5)$
- domain and range; D: $(-\infty,\infty)$ R: $[3,\infty)$
- maximum or minimum value.

minimum is at vertex. minimum at $x=-1$ so $y=3$

no max for this one. $y=3$ is minimum.

Vertex Form and Shifts

In vertex form,

$$\boxed{f(x) = a(x-h)^2 + k},$$

the vertex is (h, k) and the axis of symmetry is $x = h$.

Starting with $y = x^2$:

- h gives the horizontal shift;
- k gives the vertical shift;
- the sign of a gives the opening direction;
- $|a|$ controls vertical stretch or compression.

• $f(x) = ax^2 + bx + c \Rightarrow f(x) = a(x-h)^2 + k$

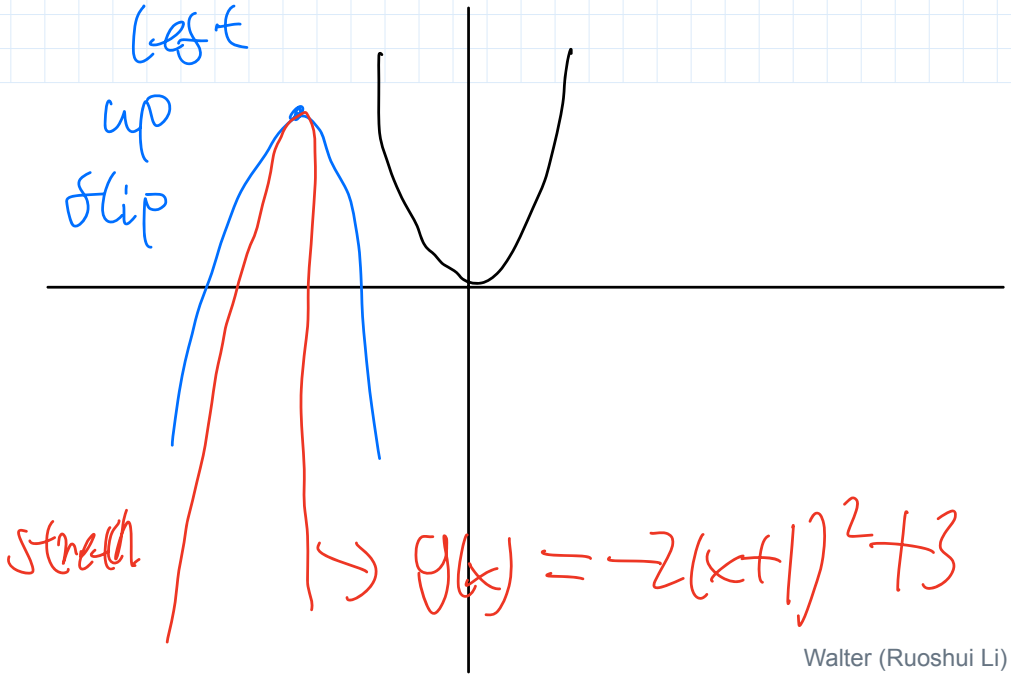
• (h, k) is vertex

Describe and sketch $g(x) = -2(x+1)^2 + 3$.

$x^2 \xrightarrow[\text{hor...}]{\text{left 1}} (x+1)^2 \xrightarrow[\text{Ver...}]{\text{up 3}} (x+1)^2+3 \xrightarrow[x\text{-axis}]{\text{flip}} -(x+1)^2+3$

$\xrightarrow[\text{by 2.}]{\text{stretch}} -2(x+1)^2+3$

Vertically,



Build a Quadratic Function

If the vertex (h, k) and another point (x_1, y_1) are known, begin with

$$f(x) = a(x - h)^2 + k.$$

Substitute the known point to solve for a.

Example. Find the quadratic function with vertex $(1, -5)$ and y-intercept -3.

1. Substitute $(h, k) = (1, -5)$ into vertex form.
2. Use the point $(0, -3)$ to find a.
3. Write the final function and verify the point.

1. $f(x) = a(x-h)^2 + k$

use $(h,k) = (1,-5)$

$f(x) = a(x-1)^2 - 5$

$(0,-3)$ is y-intercept,

plug 0 into $f(x)$, output -3,

$f(0) = -3$

$a(0-1)^2 - 5 = -3$

$a - 5 = -3$

$a = 2$

use $a=2$,

$f(x) = 2(x-1)^2 - 5$

Maximum and Minimum Values

$f(x) = ax^2 + bx + c$

$a > 0$, $\cup$

(h, k)

The vertex gives the extreme value of a quadratic function.

Condition	Extreme value	Where it occurs
$a > 0$	minimum value k	at $x = h$
$a < 0$	maximum value k	at $x = h$

For $P(t) = -5t^2 + 40t + 12$:

(a) Does the graph have a maximum or minimum?

(b) At what value of t does it occur?

(a) $a = -5 < 0$, $\cap$ maximum,

(b) $\left(-\frac{b}{2a}, P\left(\frac{-b}{2a}\right)\right)$ $a = -5$, $b = 40$, $c = 12$

$\frac{-b}{2a} = \frac{-40}{2 \cdot (-5)} = 4$

$= (4, P(4))$

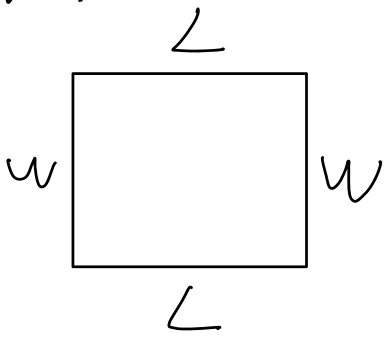
$= (4, q_2)$

at $t = 4$, $P(t)$ is maximum,

it is q_2.

Optimization: Enclose the Most Area

David has 400 yards of fencing for a rectangular field. Let w be the width. L is length.



(a) Express the length in terms of w.

(b) Write the area $A(w)$ as a quadratic function.

(c) Find the value of w that gives the largest area.

(d) Find the corresponding length and maximum area.

(e) State a reasonable domain for the model.

(a) $w+w+L+L=400$

$2w+2L=400$

$2(w+L)=400$

$w+L=200$

$L=200-w$

b). $A(w)=w\cdot L=w\cdot(200-w)=200w-w^2=-w^2+200w$

c). $-1=a$, $a<0$ $\cap$, $\frac{-b}{2a}=\frac{-200}{2(-1)}=100$

(d) $L=200-w=200-100=100$

so $100\cdot 100=10{,}000$

e). $0<w<200$